

Multimedia Computing – Assignment 2

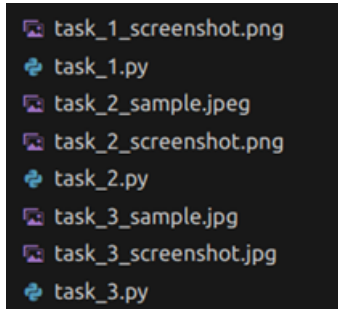
In this assignment, you will try various image processing techniques.

This assignment consists of 3 tasks.

Screenshot - For each task you need to show your resulting image like I showed for Task 1 and take a screenshot of the entire desktop window, not just the frame of the application, including the time in the task bar and everything.

Not following this instruction will get you deducted penalty points.

Naming - for each of the tasks name the images and code files just like below.



task_1.py

task_1_screenshot.png

task_2.py

task_2_screenshot.png

task_2_source.png

task_3.py

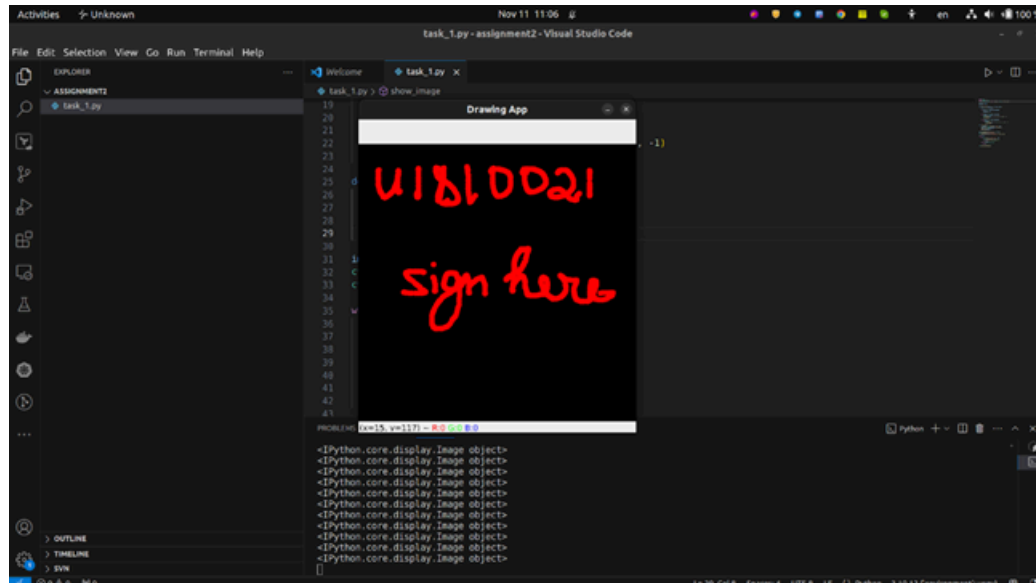
task_3_screenshot.png

task_3_source.png

where the image with “screenshot” in the name must be the final result screenshot explained above and “source” must be the photo you used as a source image for your program.

Task 1. Simple Drawing App using OpenCV

Create a simple drawing application using OpenCV in Python. The app should allow users to draw on a blank canvas by clicking and dragging the mouse.



Draw your student ID and signature and take a screenshot satisfying the requirements.

Code python file and resulting screenshot must meet all the requirements explained earlier.

Task 2. Image Segmentation using K-means Clustering Algorithm

Implement an image segmentation task using the K-means clustering algorithm. This task involves partitioning an image into a specified number of segments (k) based on the color information. K-means clustering groups similar pixels together, resulting in distinct segments.

You should get a result like below



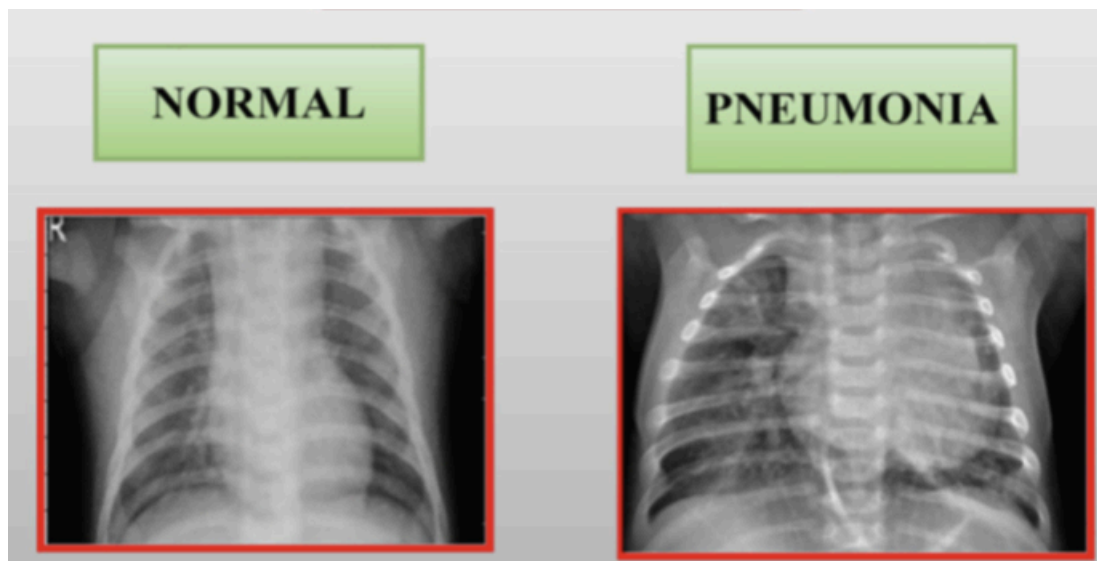
Print out the result and take a screenshot of the entire screen.

Code python file and resulting screenshot must meet all the requirements explained earlier.

Task 3. Detect Pneumonia from chest x-ray image

Use machine learning algorithm and OpenCV to detect pneumonia from an image. You should show the **detection** and **accuracy** result. The higher the accuracy is better.

You should get a result like below.



Dataset: [<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>]

Code python file with proper comment /colab file and detected pneumonia test image need to submit. (no need to submit the dataset)

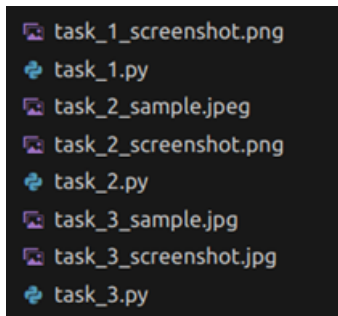
Requirements

Use your own image.

Do NOT use the exact same images shown as example above.

Each student must have a unique image for each task. If any 2 students have the same image, code it is considered cheating.

If by any chance, with or without your involvement, anybody has the same image as you, you both get 0. Therefore, it is better to use your own picture except task-3.



What to submit

Write your code and output your screenshots using VScode or any other IDE and at the end you must have your files like shown here.

Do **NOT** create folders and do **NOT** zip.

Submit these 8 files.

Python Environment.

Your code should run in the following environment.

Using python version Python 3.12.4

With packages in requirements.txt

opencv-python==4.10.0.84

numpy>=1.26.4

 Deadline 07.12.2025, 23.55 - December, Sunday, 11.55 p.m.

Late submission - below are relative to midnight

- Up to 3 hours late: 2 points will be deducted.
- Next 3 hours: 4 points will be deducted.
- Up to 24 hours: 6 points will be deducted.
- Beyond 24 hours: No submissions will be accepted, and you will receive a score of 0.

Grading

Not following the submission file type(zipping), wrong naming and not following instructions will result in a penalty.

Task 1	Working Code	2
	Result image	2
Task 2	Working Code	2
	Result image	2
Task 3	Working Code	3
	Result image and accuracy	4
Penalty	Not following naming convention	-2 (for each case)
	Wrong type file submission	
Total mark		15